NASA SPACE APPS CHALLENGE 2026 • SCIENTIFIC EVIDENCE DOSSIER

NASA Earth Science Data Inventory

Complete Technical Registry of Datasets, Variables, Scientific Rationale & Official Endpoints

Evaluation Standard: This document certifies the provenance and implementation of all 6 NASA Earth Science datasets and APIs integrated into *Climate Lens — Bangladesh*. Every parameter is mapped directly to its official NASA portal and verified for open-science reproducibility.

Platform: **Climate Lens — Bangladesh** Author: **Pavel Hasan Joy** NASA Sources: **6 Core Datasets / APIs** Time Horizon: **2001–2025 (Historical) · 2040/2050 (CMIP6)**

1. NASA SOURCES OVERVIEW TABLE

Multi-Mission Architecture

#	Dataset / Mission	NASA Center / Program	Variables Extracted	Spatial / Temporal Resolution	Direct Source Link
1	**NASA POWER API**	NASA Langley Research Center (LaRC)	`PRECTOTCORR`, `T2M_MAX`, `GWETROOT`	0.5° × 0.5° (~50 km grid)	`https://power.larc.nasa.gov/`
2	**NASA NEX-GDDP-CMIP6 Downscaled Climate Projections**	NASA Ames Research Center & NASA Center for Climate Simulation (NCCS)	`tasmax`, `pr`, `Ensemble`, `Scenarios`	0.25° × 0.25° (~27 km high resolution)	`https://www.nccs.nasa.gov/services/data-collections/land-based-products/nex-gddp-cmip6`
3	**NASA GIBS**	NASA Earth Science Data and Information System (ESDIS) Project	`Black Marble`, `Blue Marble`, `VIIRS Daily`, `GPM IMERG`	WMTS Tile Matrix (EPSG:3857)	`https://earthdata.nasa.gov/gibs`
4	**NASA SEDAC GPWv4.11**	NASA Socioeconomic Data and Applications Center (SEDAC) / CIESIN Columbia University	`PopCount`, `PopDensity`, `Calibration`	30 arc-seconds (~1 km grid)	`https://sedac.ciesin.columbia.edu/data/collection/gpw-v4`
5	**NASA MODIS NDVI**	NASA LP DAAC / USGS EROS Center	`NDVI`, `Sensor`, `Agronomics`	0.05° (~5.6 km) monthly global composite	`https://lpdaac.usgs.gov/products/mod13c2v006/`
6	**NASA EONET v3**	NASA Goddard Space Flight Center (GSFC)	`Floods`, `Severe Storms`, `Landslides`, `Wildfires`	Real-time programmatic REST API	`https://eonet.gsfc.nasa.gov/`

2. IN-DEPTH SCIENTIFIC BREAKDOWN OF NASA SOURCES

Methodology & Rationale

SOURCE 01 NASA POWER API (Prediction Of Worldwide Energy Resources)

Primary Meteorological Backbone

NASA Langley Research Center (LaRC), Climate Science Branch

Official Portal: `https://power.larc.nasa.gov/`

Endpoint: `https://power.larc.nasa.gov/api/temporal/`

Resolution: 0.5° × 0.5° (~50 km grid); 2001–2025 monthly + rolling ~400-day daily series

Access: Public Domain (NASA Open Data Policy)

`PRECTOTCORR`: Precipitation Corrected (daily mm/day and monthly cumulative mm)

`T2M_MAX`: Maximum 2-Meter Air Temperature (°C)

`GWETROOT`: Root-Zone Soil Wetness (0.0 to 1.0 fractional saturation across 0–100 cm depth)

Why This NASA Source Was Selected: Ground-based weather stations in Bangladesh frequently suffer from spatial coverage gaps, reporting discontinuities, and closed proprietary archives. NASA POWER provides continuous, satellite-derived daily and monthly meteorology assimilated with MERRA-2 and GEOS models, providing an unbroken 25-year empirical record across all 64 district centroids.

Implementation in Climate Lens: Forms the core baseline (2001–2010 multi-year average) and current observation layer ('Now'). Drives non-parametric Theil–Sen decadal slope estimation and Mann–Kendall significance testing ($p < 0.05$). Calculates extreme weather indicators (Heatwave Days where T_max >= 36°C, Consecutive Dry Spells, and Heavy Rain Days >= 50 mm/day). Queried dynamically in-browser via useLiveDaily.ts with Web Worker processing.

SOURCE 02 NASA NEX-GDDP-CMIP6 Downscaled Climate Projections

Physical Climate Model Ensembles

NASA Ames Research Center & NASA Center for Climate Simulation (NCCS)

Official Portal: `https://www.nccs.nasa.gov/services/data-collections/land-based-products/nex-gddp-cmip6`

Endpoint: `https://portal.nccs.nasa.gov/datashare/NEX-GDDP-CMIP6/`

Resolution: 0.25° × 0.25° (~27 km high resolution); 2015–2050 timeline

Access: Public Domain (NASA Open Data Policy)

`tasmax`: Daily Maximum Near-Surface Air Temperature (°C)

`pr`: Precipitation Flux (mm/day)

`Ensemble`: 5 GCMs: GFDL-ESM4, MPI-ESM1-2-HR, MRI-ESM2-0, EC-Earth3, UKESM1-0-LL

`Scenarios`: SSP2-4.5 (Moderate Mitigation) and SSP5-8.5 (High Emissions)

Why This NASA Source Was Selected: Global General Circulation Models (GCMs) typically operate at coarse grids (~100–250 km), failing to resolve Bangladesh's regional microclimates (such as the Barind tract semi-arid zone vs northeastern Sylhet wetlands). NASA NEX-GDDP applies daily statistical Bias-Correction and Spatial Disaggregation (BCSD) downscaling to 0.25°, enabling rigorous, localized mid-century projections.

Implementation in Climate Lens: Powers the static Future horizon (2040 IPCC adaptation benchmark) and continuous 2001→2050 timeline animation. Renders the multi-model median alongside a 10th-to-90th percentile uncertainty ribbon in YearChart.tsx directly adjacent to the empirical Theil–Sen statistical trend line, enabling instant comparison between historical momentum and coupled physical climate models.

SOURCE 03 NASA GIBS (Global Imagery Browse Services)

Satellite Earth Observation Basemaps

NASA Earth Science Data and Information System (ESDIS) Project

Official Portal: https://earthdata.nasa.gov/gibs

Endpoint: https://gibs.earthdata.nasa.gov/wmts/epsg3857/best/

Resolution: WMTS Tile Matrix (EPSG:3857); native sensor resolutions (500m to 10 km)

Access: Public Domain (NASA Open Data Policy)

- **Black Marble**: VIIRS Day/Night Band 500m Nighttime Lights showing urban electrification
- **Blue Marble**: 500m true-color global surface reflectance and topography
- **VIIRS Daily**: Near-real-time daytime surface reflectance and cloud systems
- **GPM IMERG**: Global Precipitation Measurement 30-minute rain radar overlay

Why This NASA Source Was Selected: Abstract vector maps disconnect decision-makers from geographical realities. NASA GIBS delivers standardized, high-speed, full-resolution satellite imagery tiles that contextualize vector choropleths with actual night lights, terrain topography, vegetation cover, and active storm clouds.

Implementation in Climate Lens: Integrated into MapLibre GL 3D (mapStyle.ts) as dynamic raster tile layers positioned directly beneath semi-transparent 3D district extrusions. Includes an interactive basemap switcher and real-time opacity slider controls.

SOURCE 04 NASA SEDAC GPWv4.11 (Gridded Population of the World)

Demographic Risk & Climate Exposure Modeling

NASA Socioeconomic Data and Applications Center (SEDAC) / CIESIN Columbia University

Official Portal: https://sedac.ciesin.columbia.edu/data/collection/gpw-v4

Endpoint: https://sedac.ciesin.columbia.edu/data/set/gpw-v4-population-count-rev11

Resolution: 30 arc-seconds (~1 km grid); calibrated to BBS 2022 Census

Access: Public Domain (NASA Open Data Policy)

- **PopCount**: Gridded population count allocated proportionally to administrative units
- **PopDensity**: Human population density per square kilometer
- **Calibration**: Calibrated to Bangladesh Bureau of Statistics (BBS) 2022 Census

Why This NASA Source Was Selected: Climate hazards are meaningful only when contextualized by human exposure. Unpopulated wetlands and dense urban clusters cannot be weighted equally. NASA SEDAC provides high-resolution gridded demographics enabling accurate calculations of affected human populations.

Implementation in Climate Lens: Powers PopulationImpactCard.tsx inside the District Details panel. Intersects demographic density with statistically verified climate signals ($p < 0.05$), revealing that 162.3 Million citizens (98.4% of Bangladesh's population) live in districts experiencing statistically significant monsoon wetting surges and flash flooding.

SOURCE 05 NASA MODIS NDVI (Vegetation Health Index)

Remote Sensing Vegetation Vigor

NASA LP DAAC / USGS EROS Center

Official Portal: https://lpdaac.usgs.gov/products/mod13c2v006/ **Endpoint:** https://doi.org/10.5067/MODIS/MOD13C2.006

Resolution: 0.05° (~5.6 km) monthly global composite; 2001–2025 series **Access:** Public Domain (NASA Open Data Policy)

`NDVI`: Normalized Difference Vegetation Index: (NIR - Red) / (NIR + Red)

`Sensor`: Moderate Resolution Imaging Spectroradiometer (Terra/Aqua MOD13C2 / MOD13A2)

`Agronomics`: Monthly crop canopy greenness across Aus, Aman, and Boro seasons

Why This NASA Source Was Selected: Vegetation vigor provides satellite-verified empirical ground-truth of agricultural health. Tracking 25 years of monthly canopy greenness verifies whether meteorological soil moisture deficits directly translate into real vegetative stress in agricultural fields.

Implementation in Climate Lens: Powers NdviVigorCard.tsx inside the Agriculture Panel. Evaluates vegetative health across rice phenological phases and visualizes canopy anomalies during documented drought periods.

SOURCE 06 NASA EONET v3 (Earth Observatory Natural Event Tracker)

Real-Time Natural Hazard Event Tracking

NASA Goddard Space Flight Center (GSFC)

Official Portal: https://eonet.gsfc.nasa.gov/ **Endpoint:** https://eonet.gsfc.nasa.gov/api/v3/events

Resolution: Real-time programmatic REST API; bounding box filtered (87.5°E–93.0°E, 20.0°N–27.0°N) **Access:** Public Domain (NASA Open Data Policy)

`Floods`: Severe riverine and flash flooding events

`Severe Storms`: Tropical cyclones, monsoon depressions, and convective storms

`Landslides`: Monsoon-triggered hill-tract landslides

`Wildfires`: Satellite thermal anomaly detections

Why This NASA Source Was Selected: EONET aggregates curated, near-real-time natural hazard events detected by NASA Earth science satellites, providing programmatic feeds of acute disaster events rather than static retrospective catalogs.

Implementation in Climate Lens: Feeds the 'Happening Now' banner and renders active disaster pins directly onto the 3D map canvas. Automatically refreshed via useEonetFeed.ts with 1–hour client-side caching.

3. COMPLEMENTARY OPEN DATA SOURCES (MULTI-AGENCY CROSS-VALIDATION)

Ground-Truthing

Dataset / Authority	Providing Organization	Role & Integration in Platform	Source Link
geoBoundaries (ADM1 & ADM2)	William & Mary / Runfola et al.	Provides simplified vector GeoJSON boundaries for all 8 divisions and 64 districts (CC BY 4.0).	https://www.geoboundaries.org/
GBIF Occurrences API	Global Biodiversity Information Facility	Supplies 1,056 geo-referenced observation records for 6 national indicator species (Tiger, Dolphins, Hilsa, Elephant, Fishing Cat).	https://api.gbif.org/v1/
World Bank Open Data / FAOSTAT	World Bank & United Nations FAO	Supplies 24-year national crop yield time series (2000–2023) for non-parametric Spearman rank correlation analysis against detrended climate.	http://api.worldbank.org/v2/country/BGD/indicator/AG.YLD.CREL.KG
BRRI & DAE Agronomic Guidelines	Bangladesh Rice Research Institute & Department of Agricultural Extension	Establishes standardized phenological crop calendars and vulnerability thresholds for Aman, Aus, and Boro rice.	http://www.brri.gov.bd/
Peer-Reviewed Climate Impact Studies (14 Papers)	PLOS Climate, Elsevier, Springer, IUCN Red List (2015)	Curates 14 peer-reviewed scientific studies documenting direct and contributing climate impacts across agriculture, fisheries, and wildlife.	https://github.com/pavel-hasan-joy/Climate-Lens/blob/main/docs/sources.md